Investors cannot rely on long-horizon R^2s because they are *spurious*. This means that while the true predictability is very weak (we say that there is a small degree of predictability in *population*), the regression R^2s are much larger in *small samples*. Small samples here can mean hundreds or thousands of years. The intuition for this spurious effect is that the overlapping observations in equation (8.7) artificially induce dependence—the three-period long-horizon return from t to t+3 overlaps two periods with the three-period long-horizon return from t+1 to t+4—and the problem gets worse as the horizons get longer.[26] Valkanov (2003) and Boudoukh, Richardson, and Whitelaw (2008), among others, derive the correct small sample R^2 distributions and show they are hugely biased upward; using the simple R^2 always overstates, often grossly so, long-horizon predictability. The robust t-statistic corrections get the calculations right; R^2s do not. The moral of the story is that there is not that much predictability in the data—which is as it should be according to theory—and it is also hard to detect at long horizons.

Predictability Comes and Goes

A final twist makes it even harder for investors to predict returns in practice: the predictive coefficients themselves (the b parameter in equation (8.7)) vary over time. Statistical models that can accommodate this *parameter instability* do so by allowing the coefficients to undergo structural breaks or to change slowly over time.[27] Henkel, Martin, and Nardari (2011) capture regime-dependent predictability. They find that predictability is weak during business cycle expansions but very strong during recessions. Thus, predictability is counter-cyclical and is observed most during regimes of economic slowdowns. [28]

Implications for Investors

There are time-varying risk premiums, but they are difficult to estimate. If you attempt to take advantage of them, do the following:

1. Use good statistical techniques.

   Overstating statistical significance, for example, by using the wrong t-statistics and thereby making predictability look "too good," will hurt you when you implement investment strategies. One manifestation of spuriously high R^2 in fitted in samples is that the performance deteriorates markedly

[26] The literature on spurious regressions calls these *partially aggregated variables*.

[27] See, for example, Paye and Timmermann (2006) and Lettau and Van Nieuwerburgh (2008) for structural breaks in predictability relations. Dangl and Halling (2012) allow the predictive coefficient to change slowly following a random walk specification. The coefficients of predictability change in Johannes, Korteweg, and Polson (2011) as investors learn over time.

[28] For a summary of regime changes and financial markets, see Ang and Timmermann (2012).